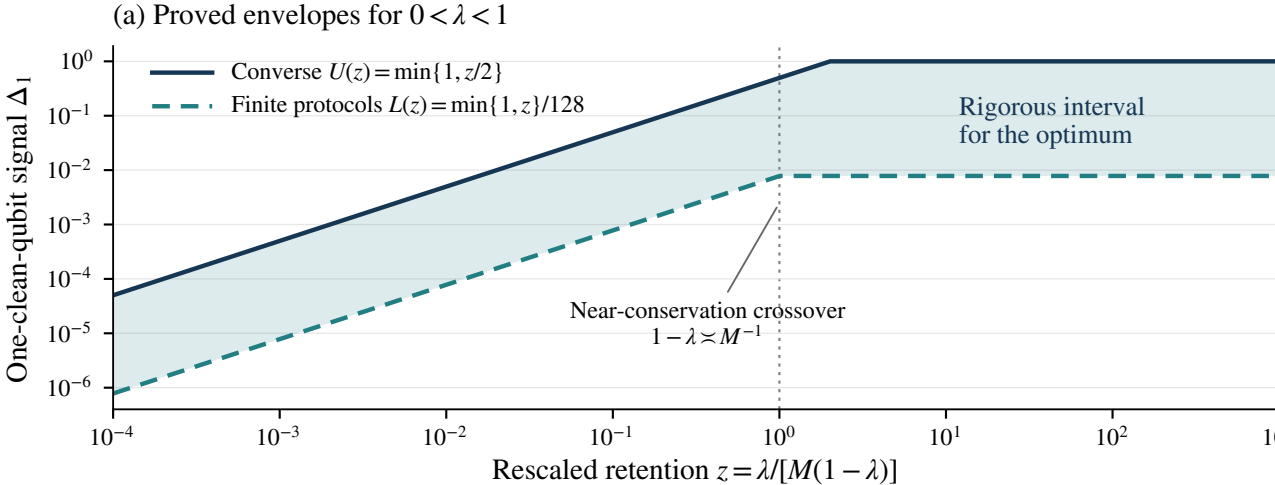
(a) Proved envelopes for $0 < \lambda < 1$
Converse $U(z) = \min\{1, z/2\}$
Finite protocols $L(z) = \min\{1, z\}/128$
Rigorous interval for the optimum
Near-conservation crossover
$1 - \lambda \asymp M^{-1}$
One-clean-qubit signal Δ_1
Rescaled retention $z = \lambda/[M(1-\lambda)]$
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}